

1. Function

- A **function** is a reusable block of code that performs a specific task.
It is generally defined outside a class.

```
def add(a, b):  
    return a + b  
  
print(add(10, 20))
```

2. Method

- A **method** is a function defined inside a class.
It is used to perform an operation related to an object or class.

```
class Student:  
    def study(self):  
        print("Student is studying")  
  
s1 = Student()  
s1.study()
```

3. Difference

Function	Method
Defined independently	Defined inside a class
Usually called directly	Usually called through an object/class
Does not require self	Instance methods normally have self
Example: add()	Example: student.display()

- Function → outside class
- Method → inside class

4. Normal Method

- A normal method is an instance method used to perform an operation or define the behavior of an object.

```
def display(self):  
    print(self.name)
```

5. __init__() / Constructor

- A constructor is a special method that is automatically called when an object is created and is commonly used to initialize the object's attributes.

```
class Student:  
  
    def __init__(self, name, age):  
        self.name = name  
        self.age = age
```

6. Constructor vs Normal Method

Constructor (__init__)	Normal Method
Special method	Regular instance method
Automatically called during object initialization	Usually called explicitly
Used mainly to initialize object state	Used to perform operations/behavior
Normally named __init__	Can have any valid method name
Called when creating an object	Called when needed
Commonly receives initial object data	Receives parameters required for its operation